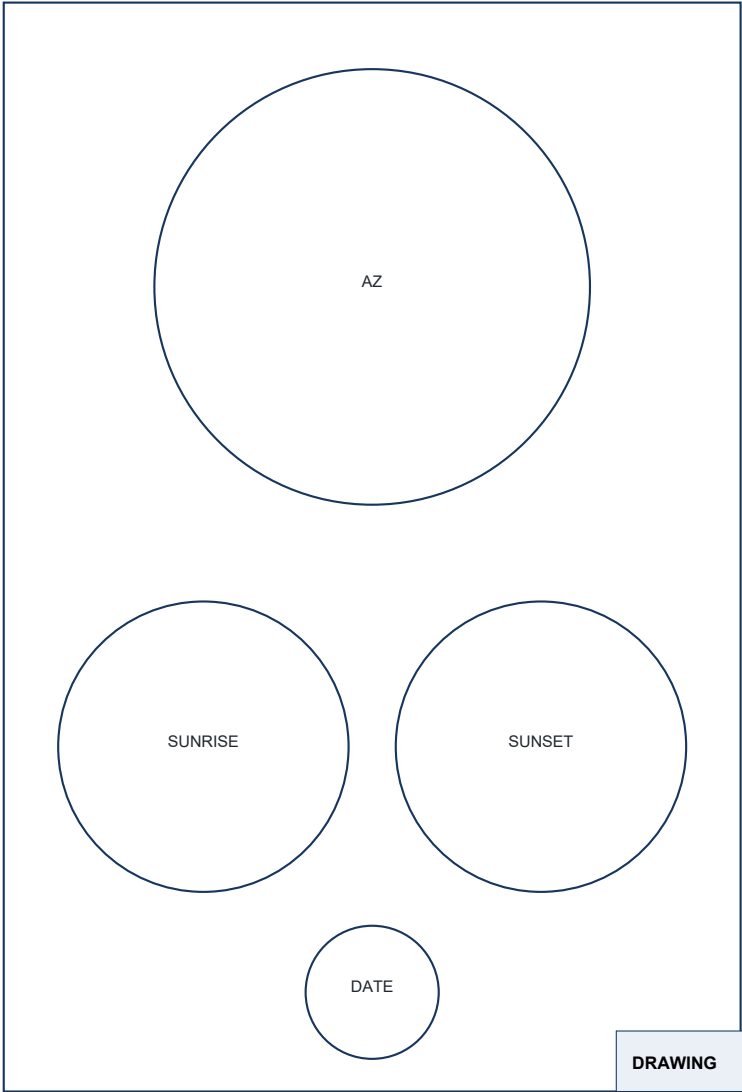


P2-A00 DRAWING INDEX

P2-A01	GENERAL ARRANGEMENT
P2-A02	EXPLODED MODULE STACK
P2-A03	KINEMATIC / INTERFACE DIAGRAM
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P2-102	SUNRISE-AZIMUTH CAM
P2-103	CAM HUB
P2-104	CAM MODULE PLATE
P2-105	DAYLIGHT SPAN LEVER
P2-106	AZIMUTH SPAN LEVER
P2-107	TIME CARRIER PLATE
P2-108	TIME MODULE PLATE
P2-109	AZIMUTH MODULE PLATE
P2-110	SUNRISE TIME SHAFT
P2-111	SUNSET OUTPUT SHAFT
P2-112	AZIMUTH INNER SHAFT
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P2-114	FRONT LAYOUT
P2-115	FRONT GUARD
P2-116	TIME DIAL
P2-117	AZIMUTH DIAL
P2-118	TIME HAND
P2-119	AZIMUTH HAND
P2-120	CAM RUNOUT COUPON
P2-121	HUB PILOT FIT COUPON
P2-122	LEVER SPAN GAUGE
P2-123	COMMON CAM / DATE SHAFT
P2-124	CAM FOLLOWER CARRIAGE
P2-125	RACK CLAMP
P2-126	CARRIER OUTPUT-RACK BRACKET
P2-127	AZIMUTH GEAR BUSHING
P2-128	DATE CONTROL SHAFT

P2-A01 GENERAL ARRANGEMENT



OVERALL ENVELOPE

609 W x 900 H x 200 D MAX

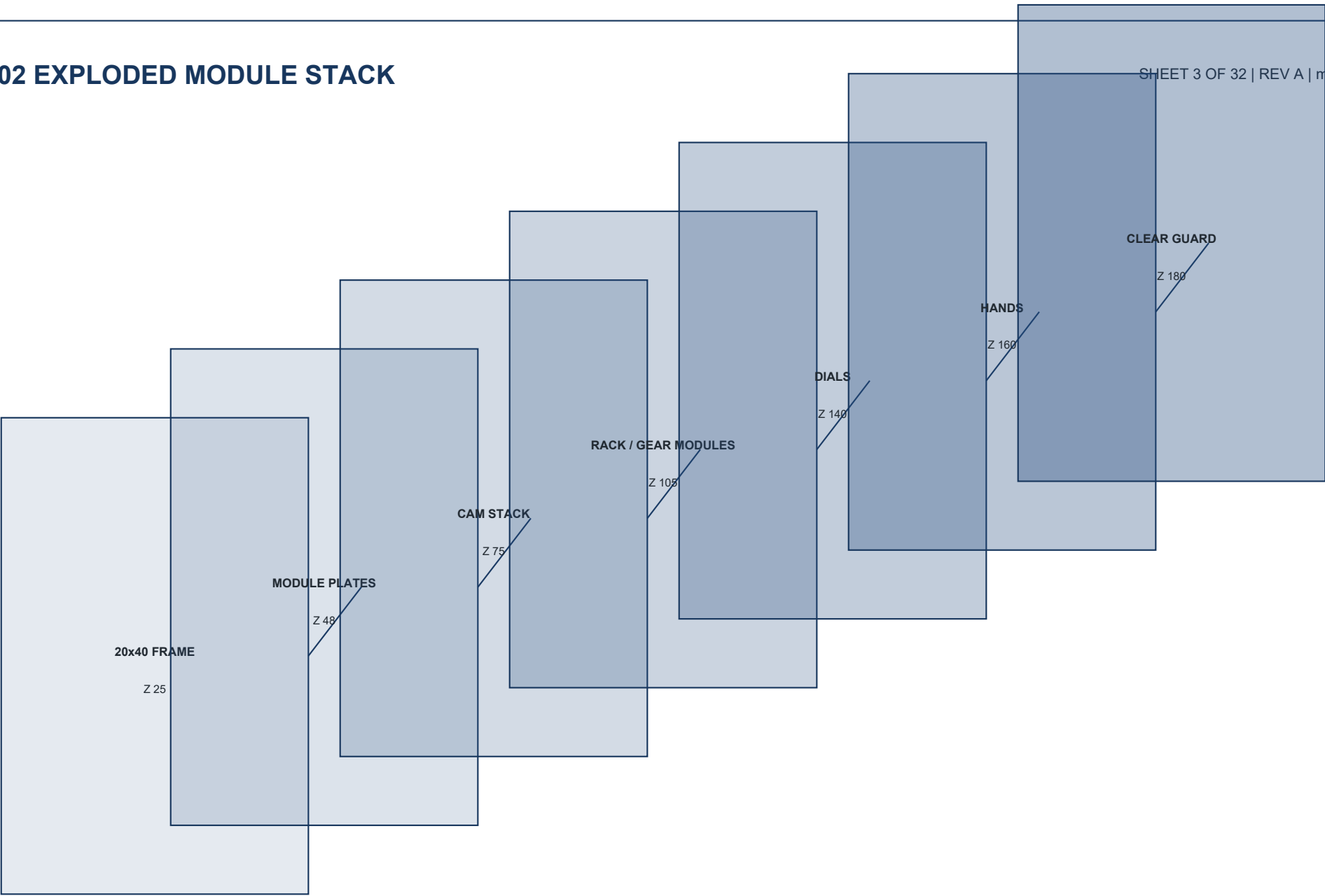
DIALS: 360 AZ / 240 TIME

GUARD INNER FACE Z174

MIN HAND/GUARD CLEARANCE 8

DRAWING	P2-A01 GENERAL ARRANGEMENT	STATUS	FOR QUOTATION
DATUM	LOWER-LEFT FRONT / Z REAR	MASTER	P2 ICD + STEP/DXF

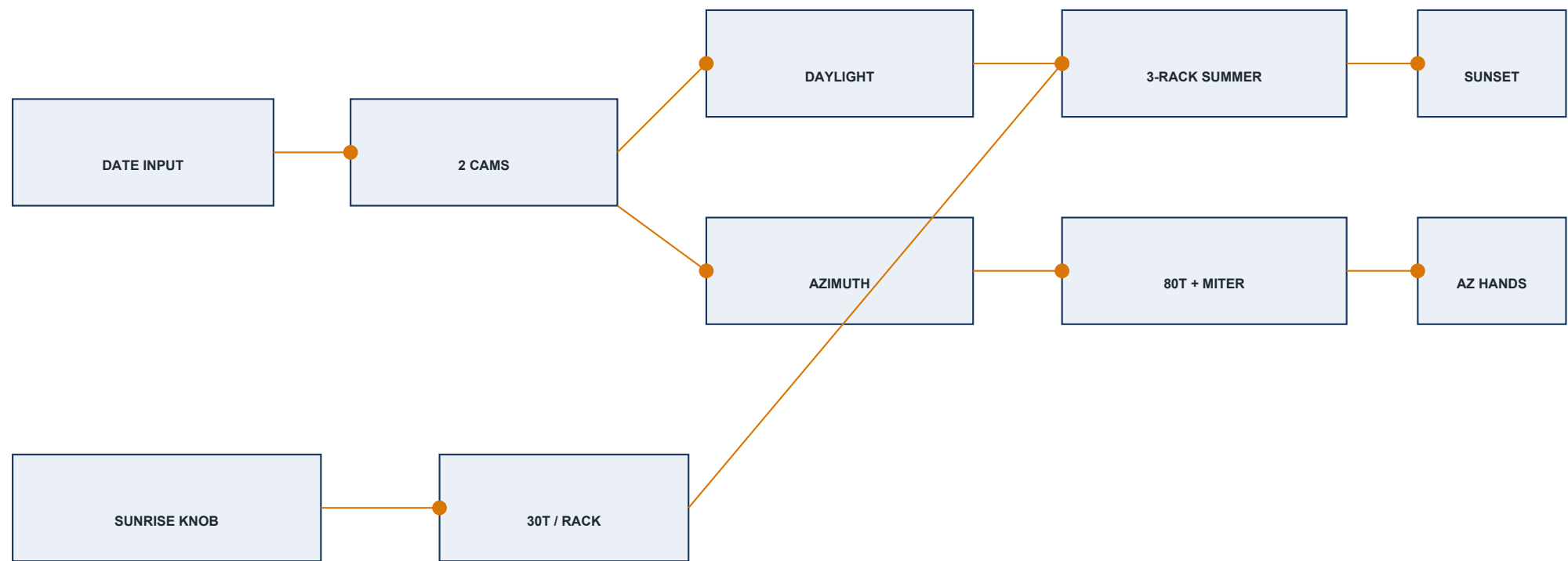
P2-A02 EXPLODED MODULE STACK



Assembly order: square frame -> datum plates -> shafts/cams -> racks/

DRAWING	P2-A02 EXPLODED MODULE STACK	STATUS	FOR QUOTATION
DATUM	LOWER-LEFT FRONT / Z REAR	MASTER	P2 ICD + STEP/DXF

P2-A03 KINEMATIC / INTERFACE DIAGRAM

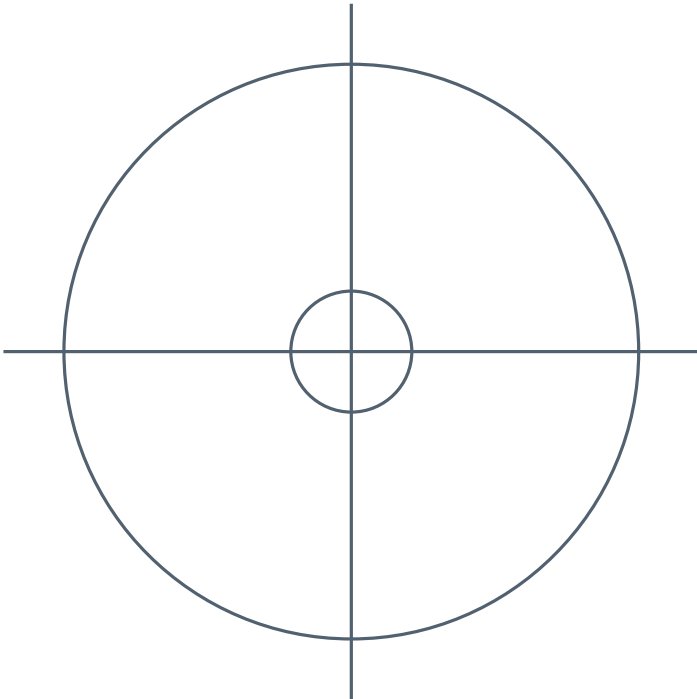


TIME: $x_{\text{carrier}} = (x_{\text{sunrise}} + x_{\text{daylight}}) / 2$; 7.5 mm output radius restores full sum.

AZIMUTH: 40 mm pitch radius gives 1 deg output per 0.6981317 mm; 1:1 miter reverses sunset.

All three modules expose independent zero/span/phase adjustments after guard removal.

DRAWING	P2-A03 KINEMATIC / INTERFACE DIAGRAM	STATUS	FOR QUOTATION
DATUM	LOWER-LEFT FRONT / Z REAR	MASTER	P2 ICD + STEP/DXF

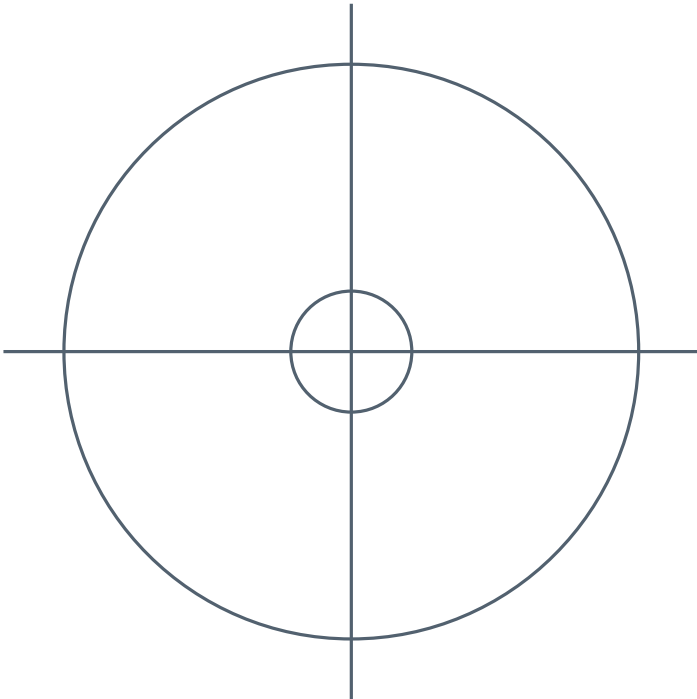


CONTROLLED CHARACTERISTICS

10.00 +/-0.03 THK
32 H7 PILOT
3x 5.5 THRU ON 40 PCD
4 H7 DATUM AT R20 / 60 DEG
PROFILE 0.05 TO A|C
Ra1.6

PART	P2-101	MATERIAL	POM-C	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

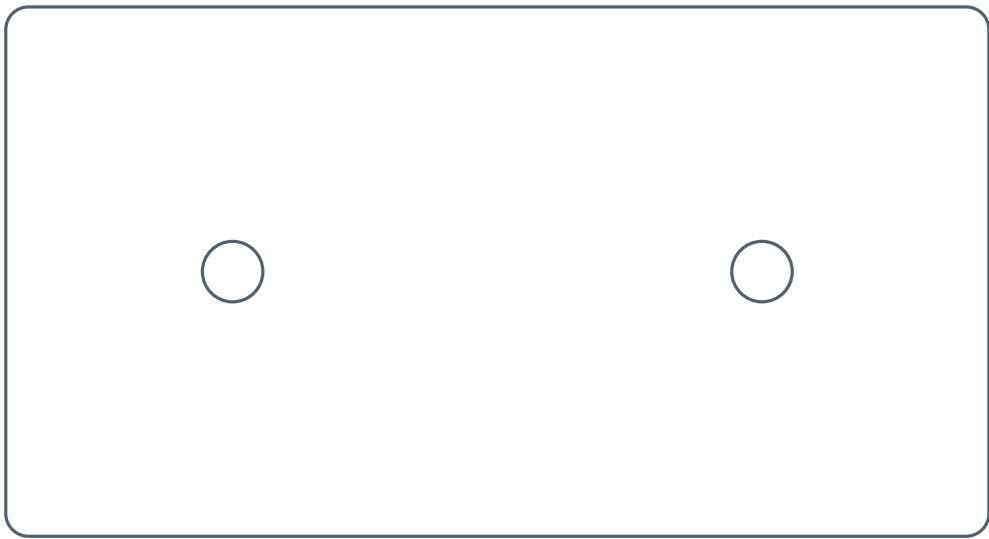


CONTROLLED CHARACTERISTICS

10.00 +/-0.03 THK
32 H7 PILOT
3x 5.5 THRU ON 40 PCD
4 H7 DATUM AT R20 / 60 DEG
PROFILE 0.05 TO A|C
Ra1.6

PART	P2-102	MATERIAL	POM-C	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.



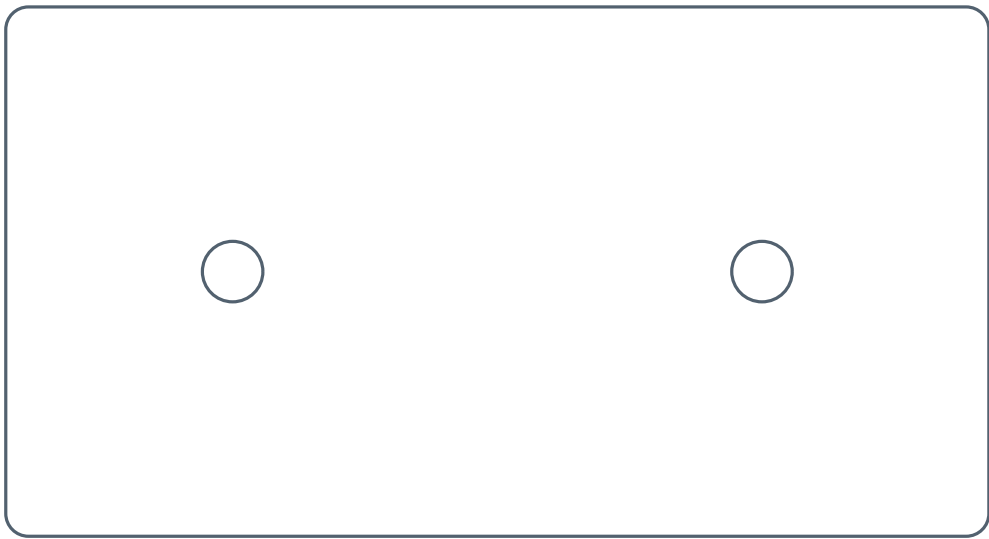
CONTROLLED CHARACTERISTICS

FLANGE OD50 x 8
PILOT 32 h6 x 3
SHAFT BORE 12 H7 KEYED
3x M5 ON 40 PCD
PILOT TIR 0.025 TO BORE

PART	P2-103	MATERIAL	6061-T6	QTY	2
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-104 CAM MODULE PLATE



CONTROLLED CHARACTERISTICS

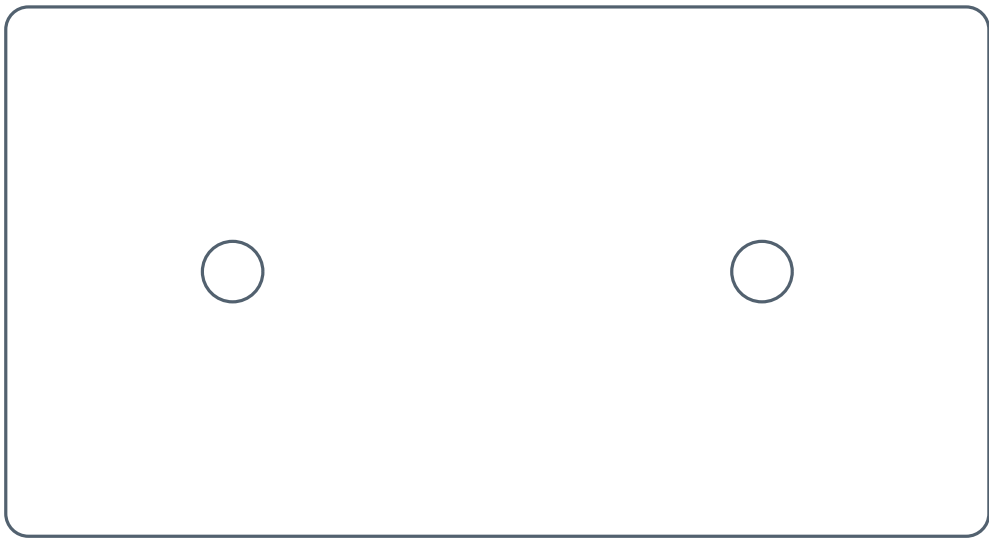
360 x 340 x 6
DATE BORE 28 AT (-80,-75)
LEVER PIVOT 8 H7 AT (80,0)
4x FRAME 6.6 AT (+/-165,+/-155)
GUIDE HOLES PER DXF

PART	P2-104	MATERIAL	6061-T6	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-105 DAYLIGHT SPAN LEVER

SHEET 9 OF 32 | REV A | mm



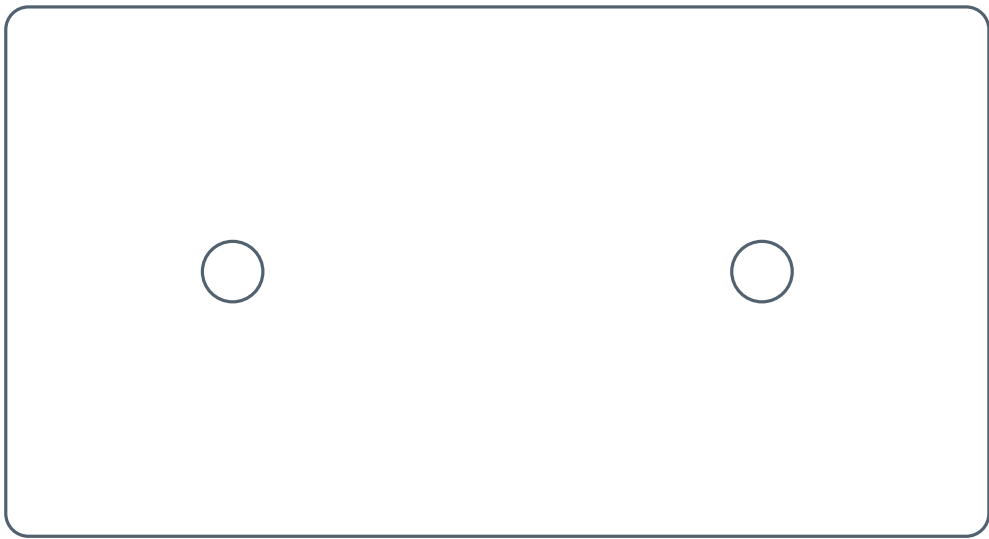
CONTROLLED CHARACTERISTICS

180 x 22 x 6
CENTER PIVOT 8 H7
2x RADIAL 6.2 x 67 SLOTS
NOMINAL GUIDE SPACING +/-75

PART	P2-105	MATERIAL	6061-T6	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-106 AZIMUTH SPAN LEVER



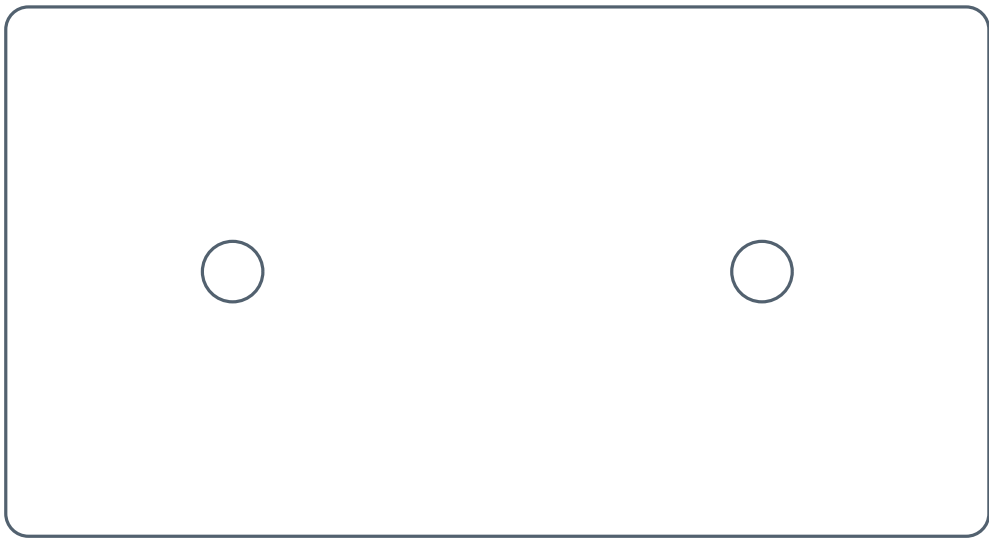
CONTROLLED CHARACTERISTICS

180 x 22 x 6
CENTER PIVOT 8 H7
2x RADIAL 6.2 x 67 SLOTS
NOMINAL GUIDE SPACING +/-75

PART	P2-106	MATERIAL	6061-T6	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-107 TIME CARRIER PLATE



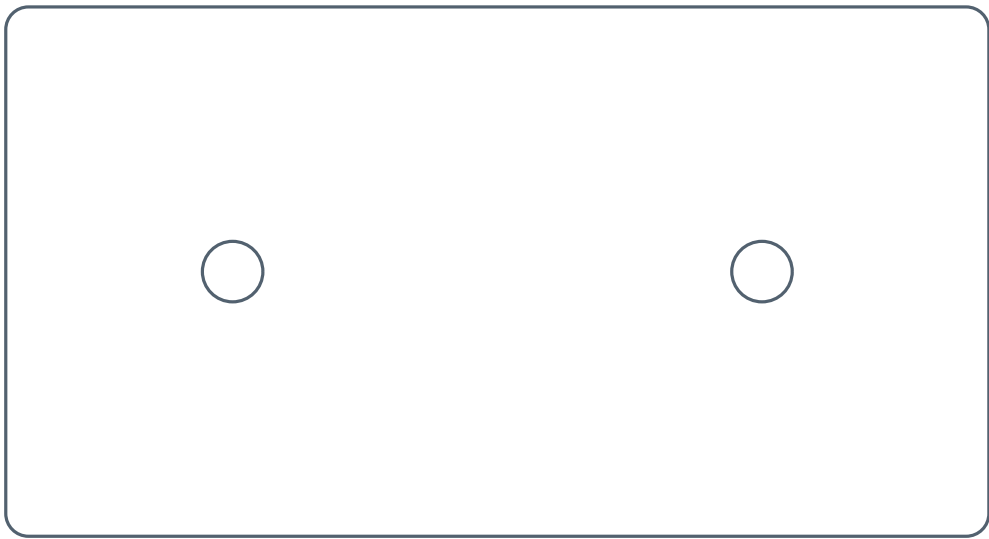
CONTROLLED CHARACTERISTICS

58 x 42 x 6
PLANET SHAFT BORE 8 H7
2x 5.2 THRU AT X +/-22

PART	P2-107	MATERIAL	6061-T6	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-108 TIME MODULE PLATE



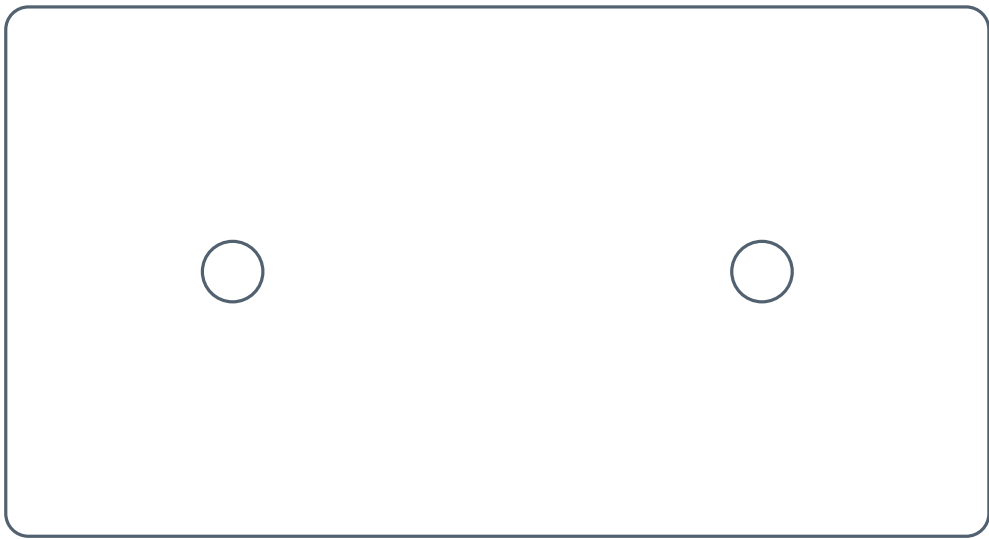
CONTROLLED CHARACTERISTICS

320 x 180 x 6
SHAFT BORES 22 AT X +/-139.5
4x FRAME 6.6 AT (+/-150,+/-75)
RAIL/STOP HOLES PER DXF

PART	P2-108	MATERIAL	6061-T6	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-109 AZIMUTH MODULE PLATE



CONTROLLED CHARACTERISTICS

190 x 130 x 6
CENTER BORE 22 H7
4x FRAME 6.6 AT (+/-80,+/-55)

PART	P2-109	MATERIAL	6061-T6	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-110 SUNRISE TIME SHAFT



CONTROLLED CHARACTERISTICS

OD8 h6 x 110
OD12 SHOULDER x 3
JOURNAL Ra0.8
TIR 0.025

PART	P2-110	MATERIAL	303 SS	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-111 SUNSET OUTPUT SHAFT



CONTROLLED CHARACTERISTICS

OD8 h6 x 95
OD12 SHOULDER x 3
JOURNAL Ra0.8
TIR 0.025

PART	P2-111	MATERIAL	303 SS	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-112 AZIMUTH INNER SHAFT



CONTROLLED CHARACTERISTICS

OD8 h6 x 150
OD12 SHOULDER x 3
JOURNAL Ra0.8
TIR 0.025

PART	P2-112	MATERIAL	303 SS	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-113 AZIMUTH OUTER TUBE



CONTROLLED CHARACTERISTICS

OD14 h8
ID8.10 +0.03/-0.00
LENGTH130
CONCENTRIC 0.025 TIR

PART	P2-113	MATERIAL	303 SS	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-114 FRONT LAYOUT



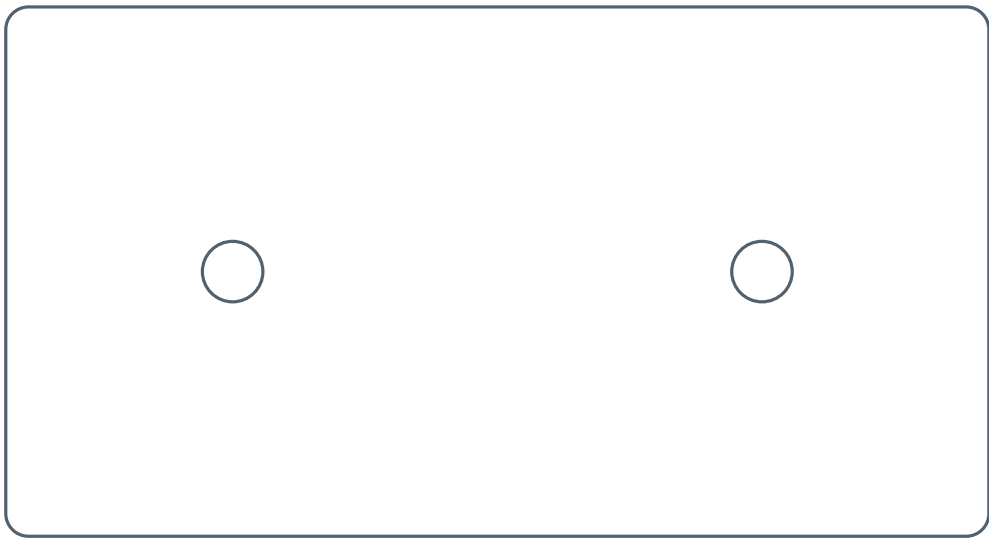
CONTROLLED CHARACTERISTICS

ENVELOPE 609 x 900
DIAL CENTERS: AZ (304.5,665), SUNRISE (165,285), SUNSET (444,285), DATE (304.5,82)

PART	P2-114	MATERIAL	REFERENCE	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-115 FRONT GUARD

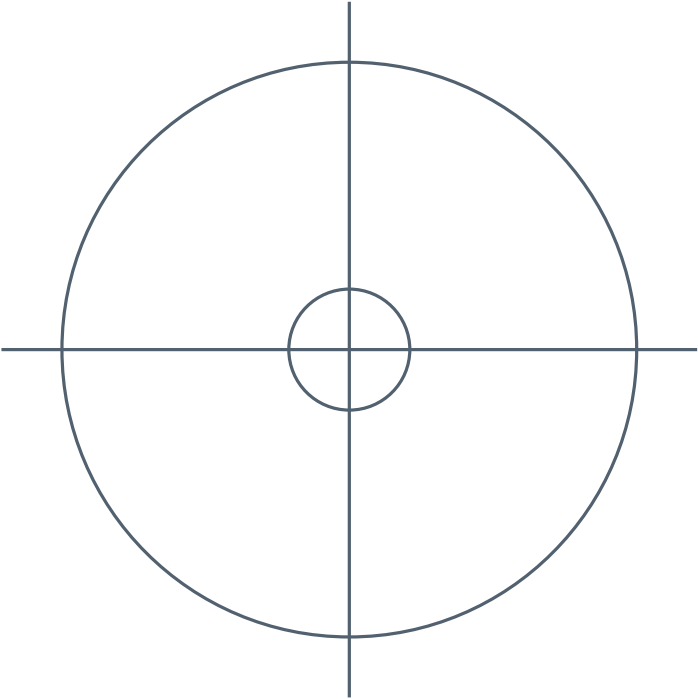


CONTROLLED CHARACTERISTICS

585 x 875 x 4.5
4x 6.6 AT (+/-286.5,+/-431)
EDGE POLISH AFTER MACHINING

PART	P2-115	MATERIAL	CLEAR POLYCARBONATE	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

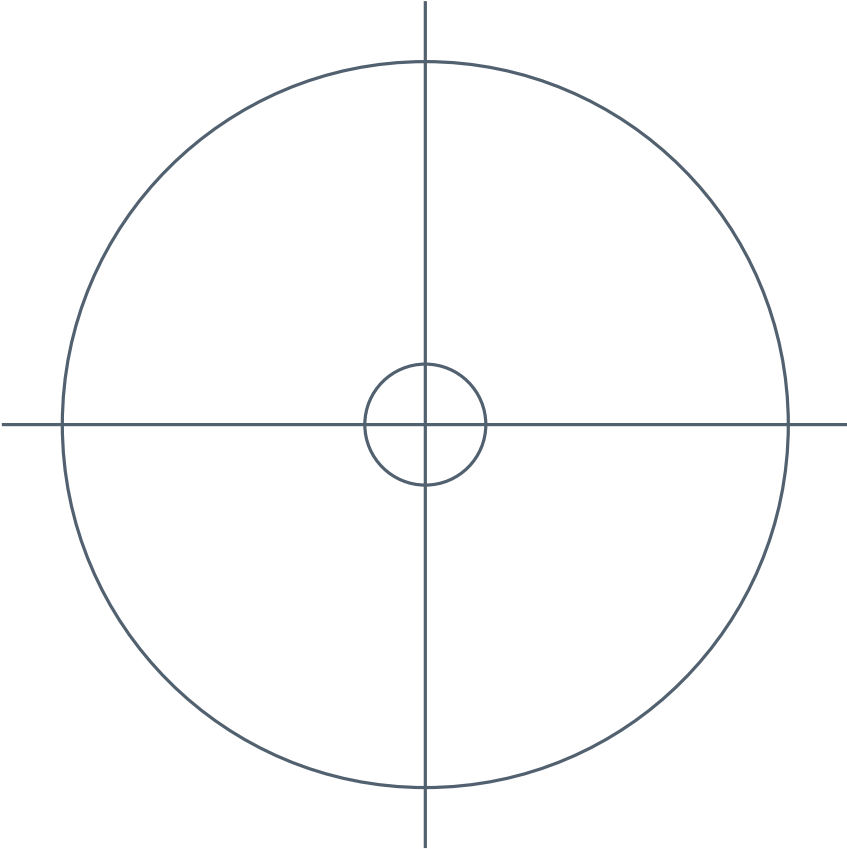


CONTROLLED CHARACTERISTICS

OD240
ID8
THK2.5
5-MIN MINOR TICKS (2.5 DEG)
30-MIN MAJOR TICKS
ARTWORK FUNCTIONAL ONLY

PART	P2-116	MATERIAL	ANODIZED ALUMINUM	QTY	2
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.



CONTROLLED CHARACTERISTICS

OD360
ID8
THK2.5
5-DEG MINOR TICKS
15-DEG MAJOR TICKS
NORTH AT +Y

PART	P2-117	MATERIAL	ANODIZED ALUMINUM	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-118 TIME HAND

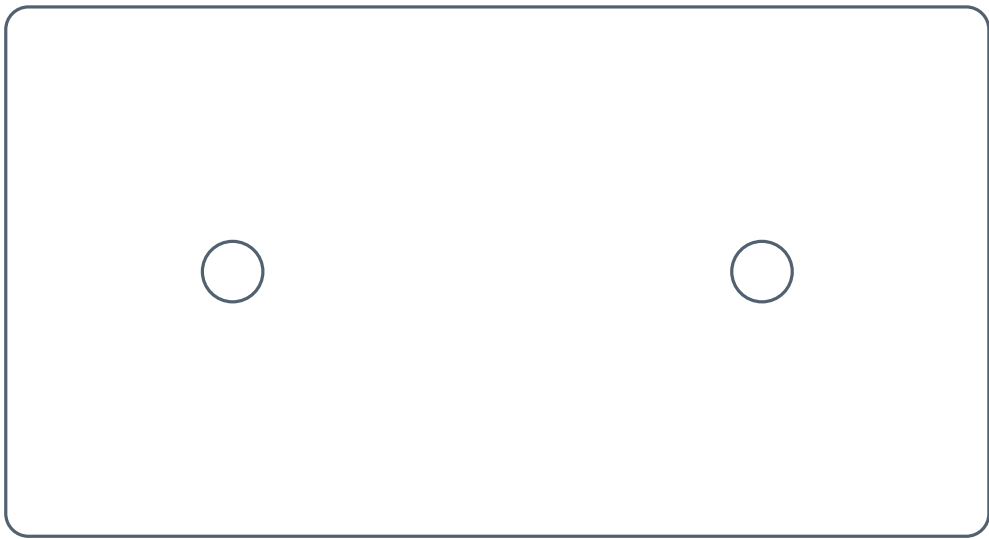


CONTROLLED CHARACTERISTICS

LENGTH105
HUB OD20
BORE8 H7
THK1.6
BALANCE AFTER FINISH

PART	P2-118	MATERIAL	5052-H32	QTY	2
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.



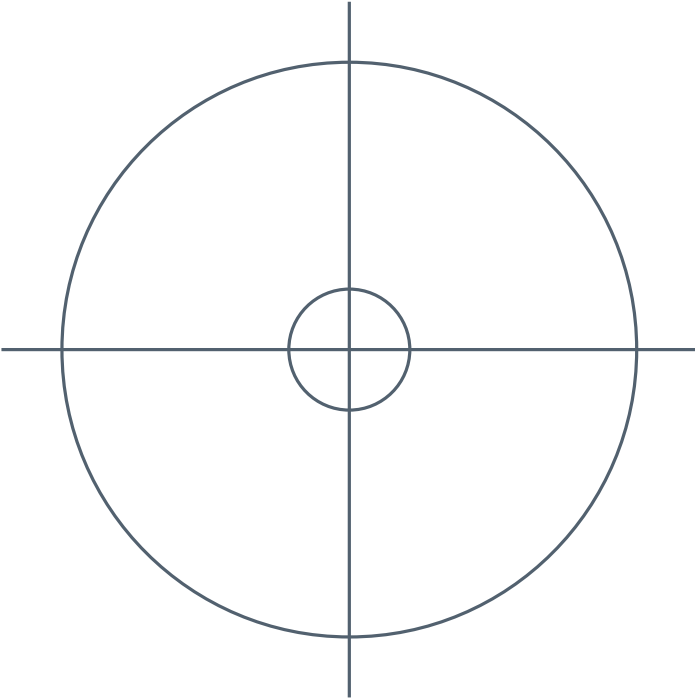
CONTROLLED CHARACTERISTICS

LENGTH165
HUB OD22
BORE8 H7
THK1.6
BALANCE AFTER FINISH

PART	P2-119	MATERIAL	5052-H32	QTY	2
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-120 CAM RUNOUT COUPON



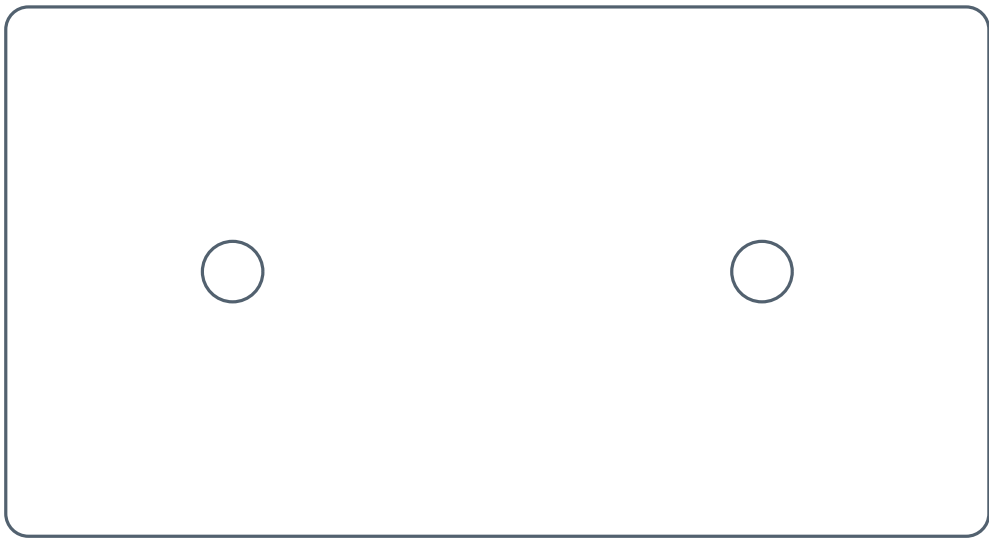
CONTROLLED CHARACTERISTICS

OD120 x 10
PRODUCTION 32 PILOT / M5 PCD / 4 DATUM
OD TIR <=0.05 ON PRODUCTION HUB

PART	P2-120	MATERIAL	POM-C	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-121 HUB PILOT FIT COUPON



CONTROLLED CHARACTERISTICS

170 x 52 x 10
BORES 31.98 / 32.00 / 32.02 / 32.04 LEFT TO RIGHT
SAME SETUP AS CAMS

PART	P2-121	MATERIAL	POM-C	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-122 LEVER SPAN GAUGE



CONTROLLED CHARACTERISTICS

125 x 18 x 6
2x 5 H7
CENTER DISTANCE 100.00 +/-0.03

PART	P2-122	MATERIAL	6061-T6	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.



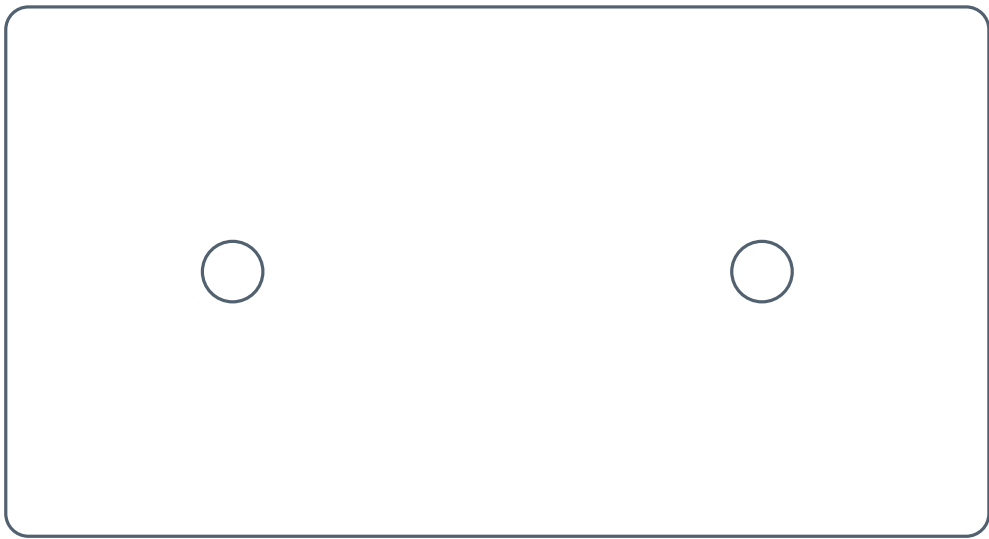
CONTROLLED CHARACTERISTICS

OD12 h6 x 160
4 WIDE x 2 DEEP KEYWAY OVER 70
JOURNAL Ra0.8
TIR 0.025

PART	P2-123	MATERIAL	303 SS	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-124 CAM FOLLOWER CARRIAGE



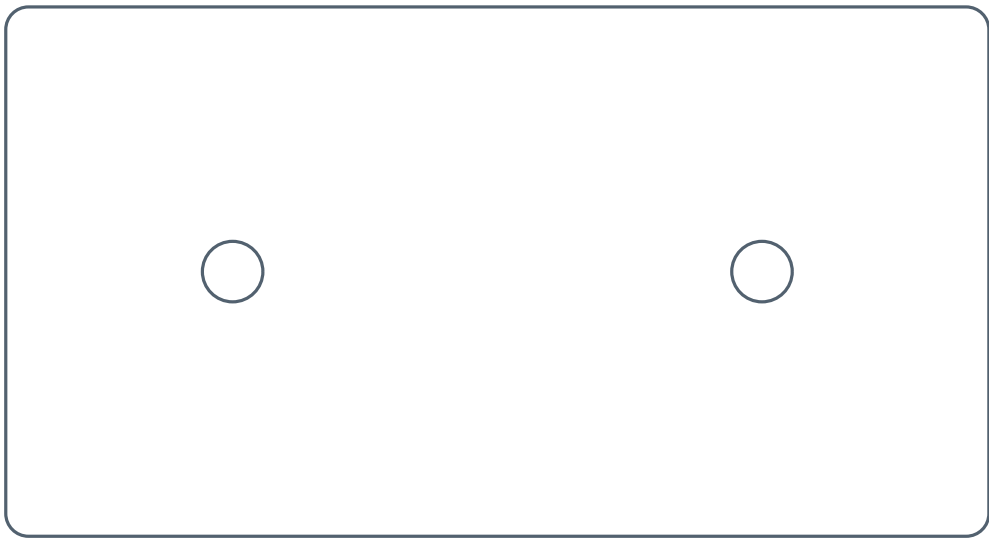
CONTROLLED CHARACTERISTICS

45 x 30 x 8
ROLLER BORE 5 H7 AT X15
2x M4 CLEARANCE AT X +/-12

PART	P2-124	MATERIAL	6061-T6	QTY	2
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-125 RACK CLAMP



CONTROLLED CHARACTERISTICS

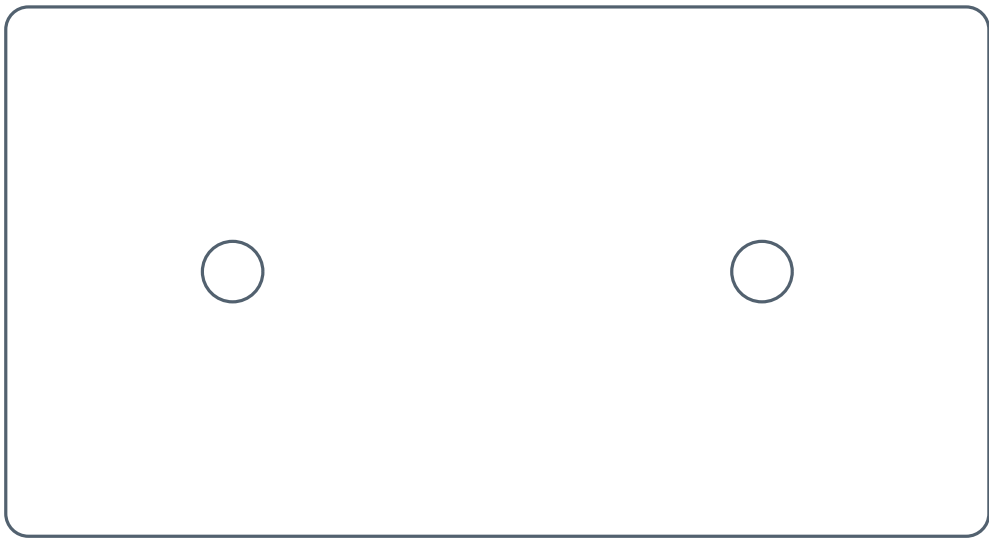
50 x 20 x 8
2x M4 CLEARANCE AT X +/-18
MATCH-DRILL TO COMMERCIAL RACK

PART	P2-125	MATERIAL	6061-T6	QTY	3
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-126 CARRIER OUTPUT-RACK BRACKET

SHEET 30 OF 32 | REV A | mm



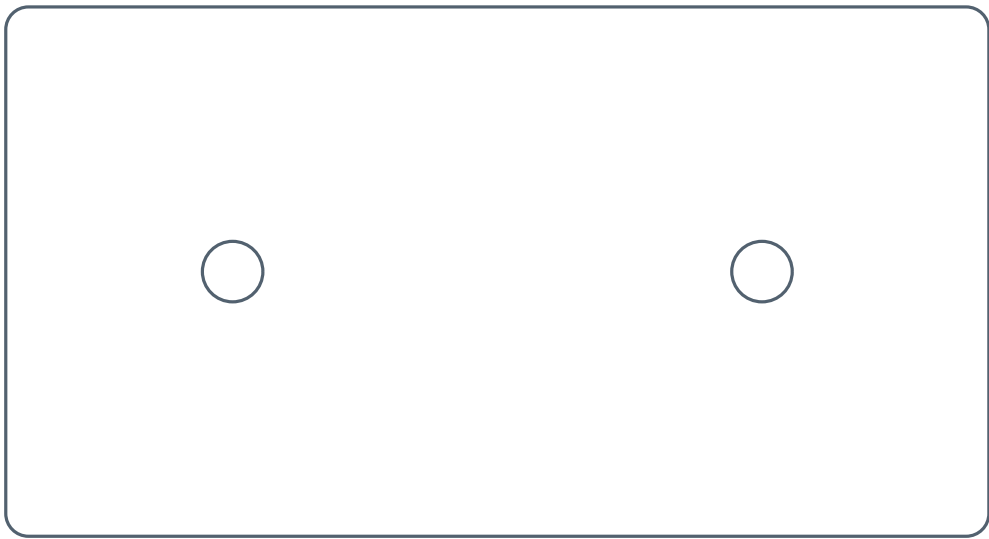
CONTROLLED CHARACTERISTICS

80 x 25 x 8
4x M4 CLEARANCE AT X +/-12,+/-30
MOUNTS THIRD RACK

PART	P2-126	MATERIAL	6061-T6	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.

P2-127 AZIMUTH GEAR BUSHING

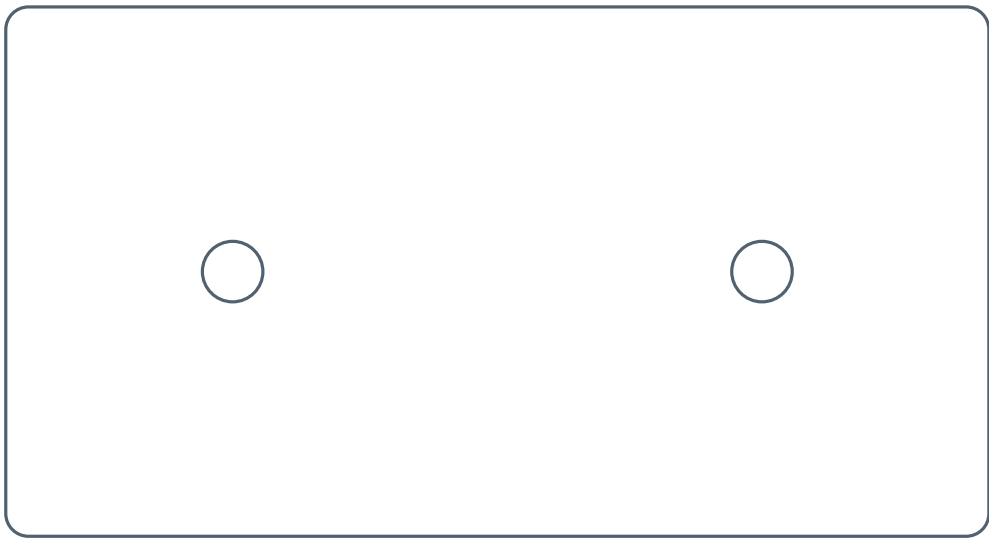


CONTROLLED CHARACTERISTICS

FLANGE OD20 x 2
BODY OD15 h6 x 10
BORE8 H7
CONCENTRIC 0.025 TIR

PART	P2-127	MATERIAL	303 SS	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.



CONTROLLED CHARACTERISTICS

OD8 h6 x 100
OD12 SHOULDER x 3
HTD PULLEY + DATE KNOB
JOURNAL Ra0.8
TIR 0.025

PART	P2-128	MATERIAL	303 SS	QTY	1
DATUMS	SEE ICD	GENERAL	ISO 2768-mK	STATUS	FOR QUOTATION
CAD	STEP/DXF MASTER	PROFILE	NO STL MACHINING	FINISH	SEE ICD

CAD defines basic geometry. PDF notes and P2 Interface Control Document define tolerances and inspection.